B2.3 Cell specialization

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b2-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 256–264 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Becoming specialized

B2.3.1 From unspecialized cells to specialized cells

pp. 256–257

After fertilization, repeated division forms unspecialized, genetically identical cells. These then **differentiate**, as different genes are expressed in each cell type. In an early embryo, **gradients** of signalling chemicals called **morphogens** show a cell its position and so affect gene expression.

B2.3.2 Properties of stem cells

p. 257

Stem cells can **divide endlessly**, and can **differentiate** along different pathways.

B2.3.3 Stem cell niches in the adult body

pp. 258–259

A **niche** is a location that can maintain stem cells, or can promote their division and differentiation. **Bone marrow** produces all types of blood cell, and **hair follicles** produce new hair and help to repair skin.

B2.3.4 Totipotent, pluripotent and multipotent

pp. 259–260

Type	Can become	Found in
Totipotent	Any cell type, including support tissues	Zygote and earliest embryo
Pluripotent	Any cell type of the body	Early embryo
Multipotent	A limited range of related cells	Adult tissue, such as bone marrow

Size and shape

B2.3.5 Cell size as part of specialization

pp. 260–262

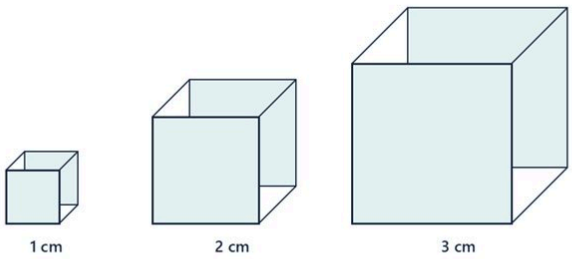
Red blood cells are small enough to pass through capillaries. Eggs are large because they store food, whereas sperm are small and able to swim. White blood cells are larger than red ones, while neurons and striated muscle fibres can be very long.

B2.3.6 Surface area-to-volume ratio limits cell size

pp. 262–264

Surface area and volume

B2.3.6



Side	SA / cm²	V / cm³	SA : V
1	6	1	6 : 1
2	24	8	3 : 1
3	54	27	2 : 1

As a cell grows, its **volume** (what needs supplies) grows faster than its **surface area** (where supplies get in).

So bigger cells have a **smaller surface area-to-volume ratio**.

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Exchange across the surface depends on **area**, whereas the need for exchange depends on **volume**. As a cell grows, volume rises faster than area, so the ratio falls. Materials then cannot enter or leave quickly enough, and the cell must stop growing or divide.

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